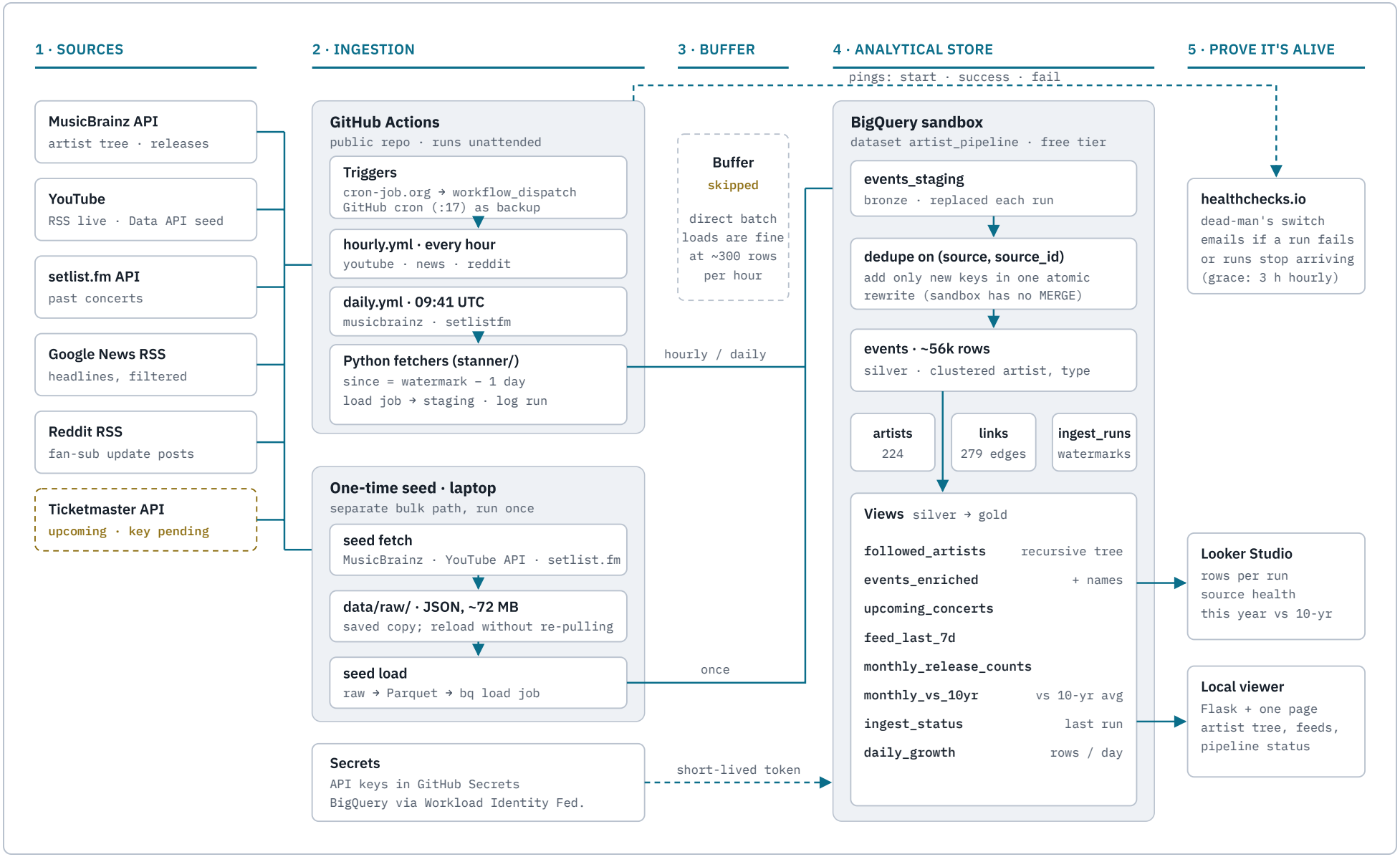
CS 452 · BUILD A PIPELINE

Stanner: artist activity pipeline

Tracks releases, official videos, concerts and news for 39 followed artists, expanded to 224 through their members and sub-units. History was bulk-loaded once; new data arrives every hour on its own.



Why each stage

STAGE	TECHNOLOGY	WHY
1 · Sources	MusicBrainz, YouTube, setlist.fm, Google News, Reddit. Ticketmaster once its key is approved.	Each covers one kind of activity: releases and the member tree, official videos and teasers, concerts, press. All are free. MusicBrainz IDs tie them together, and setlist.fm looks artists up by MusicBrainz ID directly.
2 · Ingestion	GitHub Actions running Python, triggered hourly by cron-job.org (GitHub's own cron as backup). Seed run once from a laptop.	Free for public repos, version-controlled, and runs while the laptop sleeps. GitHub's scheduler started only 3 of the first ~17 hourly slots, so an external scheduler calls the workflow-dispatch API, which GitHub runs right away. Fast feeds run hourly; MusicBrainz and setlist.fm run daily because they change slowly and MusicBrainz allows 1 request per second. The seed is a separate bulk path: pulled once, saved raw, then bulk-loaded.
3 · Buffer	Skipped	A few hundred rows an hour. Direct batch load jobs handle that without a queue.
4 · Store	BigQuery sandbox, dataset `artist_pipeline`	Free with no card: 10 GB storage and 1 TB of queries a month. About 56k rows is a few tens of MB. The sandbox has no streaming inserts or DML, so every write is a load job and dedupe is one atomic rewrite.
5 · Prove it's alive	Looker Studio, healthchecks.io and a local viewer	Looker Studio connects to BigQuery for free. It charts rows arriving per unattended run, shows each source's last run and errors, and compares this year with the 10-year monthly average. healthchecks.io is a free dead-man's switch: every run pings it, and it emails when a run fails or when pings stop, which catches a job that never starts.

Requirements

Idempotency

Each run loads into `events_staging`; only rows whose `(source, source_id)` is not already in `events` are added. Re-running the seed load added 0 rows, and the table has 0 duplicate keys.

Gaps and failures

Each source fetches from its last successful watermark minus one day. The seed's fetch time is the first watermark, which closes the seed-to-live gap. A failed run is logged as an error without moving the watermark, so the next good run backfills. Empty runs log `ok` with 0 new rows. RSS feeds only expose their latest items, so an outage longer than the feed window cannot be backfilled.

Secrets

API keys live in GitHub Secrets and a git-ignored `.env`. BigQuery access uses Workload Identity Federation, so no service-account key file exists anywhere.

Monitoring

Every run is logged in `ingest_runs` and pings healthchecks.io at start, success or failure. A failed run emails immediately; no successful ping within 3 hours (hourly) or 27 hours (daily) also emails, so a scheduler outage is caught too.

Cost

Expected $0: GitHub Actions is free for public repos, the BigQuery sandbox has no billing account, and every API, cron-job.org and healthchecks.io are on free tiers. Confirm in the GCP billing console before submitting.

As of November 13, 2026 · 56,369 events · source: docs/architecture.html